

Mathematical Models – Worked Problems

Practice: ODE-to-TF, parameter identification, and block reduction.

Ogata, *Modern Control Engineering*, Ch. 2–4 (end-of-chapter style).

Step through with **Ctrl+Enter**, or render a report with **publish**.

Contents

- Problem 1: ODE to Transfer Function
- Problem 2: Spring-Mass-Damper Parameter Identification
- Problem 3: Block-Diagram Reduction
- Problem 3 – Before vs. After Closing the Loop
- Try it yourself

Problem 1: ODE to Transfer Function

Find the transfer function $G(s) = Y(s)/U(s)$ for the system

$$y^{(3)} + 6\ddot{y} + 11\dot{y} + 6y = 6u$$

Taking the Laplace transform with zero initial conditions:

$$(s^3 + 6s^2 + 11s + 6)Y(s) = 6U(s) \quad \Rightarrow \quad G(s) = \frac{6}{s^3 + 6s^2 + 11s + 6}$$

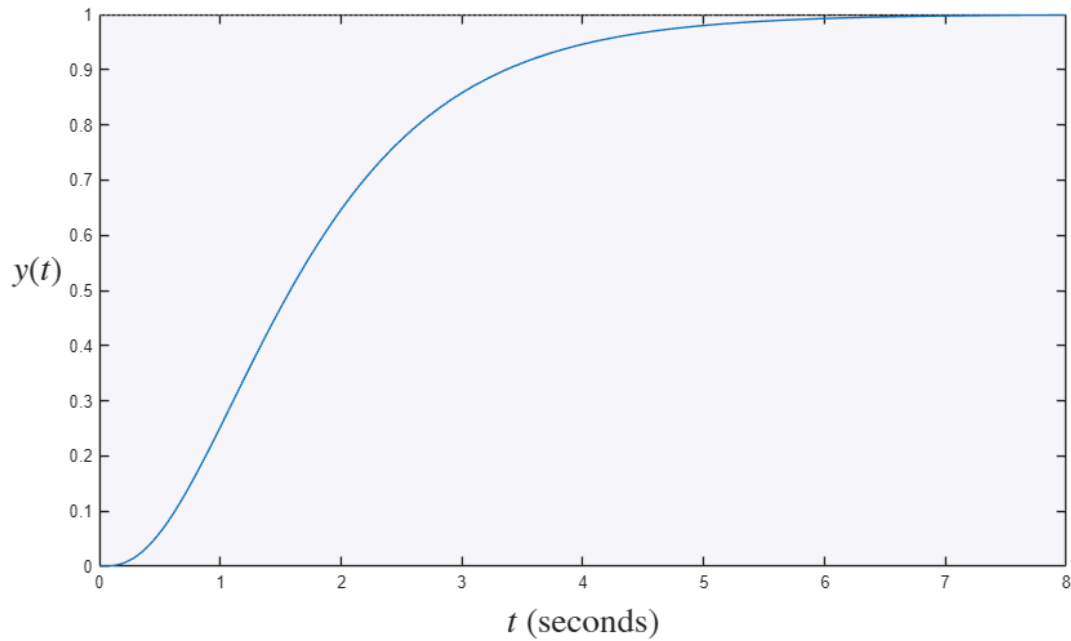
```
G1 = tf(6,[1 6 11 6]);  
poles_1 = pole(G1)
```

```
poles_1 =  
  
    -3.0000  
    -2.0000  
    -1.0000
```

The poles at $s = -1, -2, -3$ are all in the left half-plane, so the system is stable; the step response should settle without sustained oscillation.

```
figure  
step(G1)  
title('Problem 1: Step Response','Interpreter','latex','FontSize',20)  
ylabel('$y(t)$','Interpreter','latex','FontSize',20)  
set(get(gca, 'YLabel'), 'Rotation', 0)  
xlabel('$t$','Interpreter','latex','FontSize',20)
```

Problem 1: Step Response



Problem 2: Spring-Mass-Damper Parameter Identification

A mass-spring-damper system ($m\ddot{x} + b\dot{x} + kx = u$) with $m = 2$ kg is observed to have natural frequency $\omega_n = 3$ rad/s and damping ratio $\zeta = 0.5$. Recall the standard second-order form

$$\frac{1/m}{s^2 + (b/m)s + (k/m)} = \frac{1/m}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

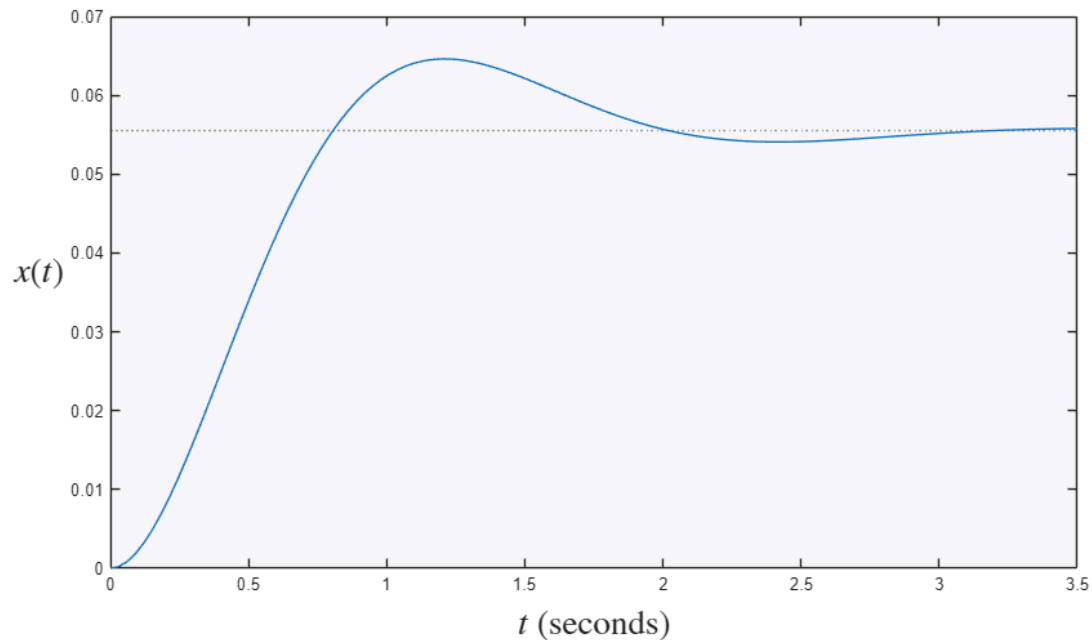
so $k = m\omega_n^2$ and $b = 2\zeta\omega_n m$.

```
m = 2; wn = 3; zeta = 0.5;
k = m*wn^2;
b = 2*zeta*wn*m;
fprintf('k = %.2f N/m, b = %.2f N.s/m\n', k, b)

G2 = tf(1/m,[1 b/m k/m]);
figure
step(G2)
title('Problem 2: Identified System Step Response','Interpreter','latex','FontSize',20)
ylabel('$x(t)$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$','Interpreter','latex','FontSize',20)

k = 18.00 N/m, b = 6.00 N.s/m
```

Problem 2: Identified System Step Response



Problem 3: Block-Diagram Reduction

Reduce a unity-feedback system whose forward path is the cascade of $G_1(s) = \frac{1}{s+2}$ and $G_2(s) = \frac{5}{s}$ (in series) to a single closed-loop transfer function, then verify its steady-state step value via the final value theorem.

```
G1 = tf(1,[1 2]);
G2 = tf(5,[1 0]);
fwd = series(G1,G2);          % forward path only (before closing the loop)
T = feedback(fwd,1)           % closed-loop (after)

ess = 1 - dcgain(T); % steady-state error for unit step (type-1 system, ess should be 0)
fprintf('Steady-state error to unit step = %.4f\n', ess)

T =

      5
-----
s^2 + 2 s + 5

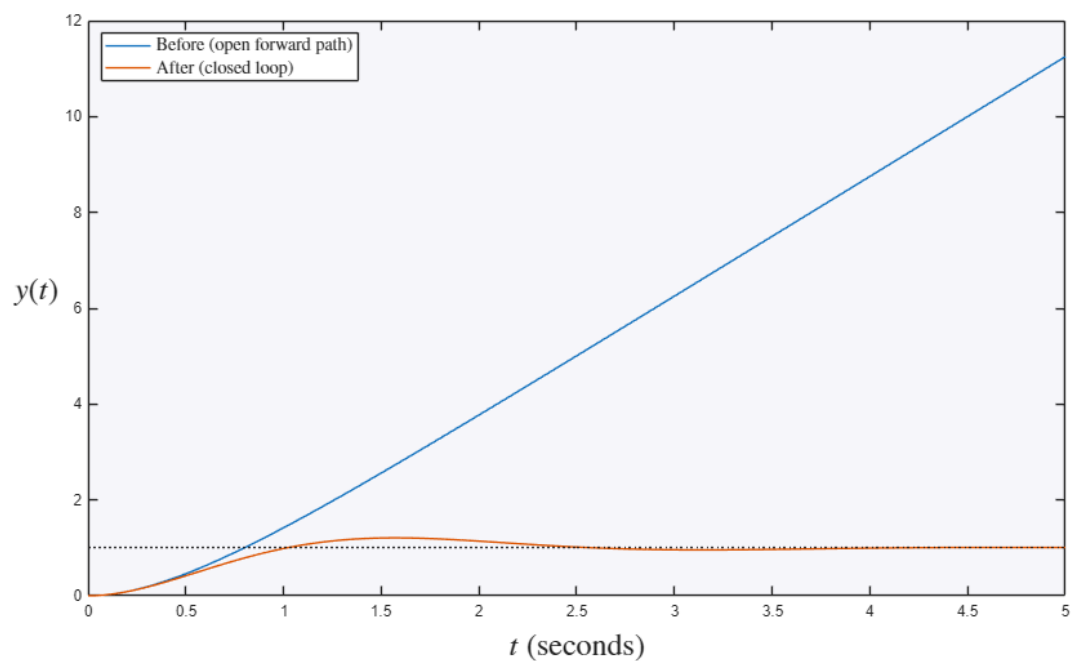
Continuous-time transfer function.
Steady-state error to unit step = 0.0000
```

Problem 3 – Before vs. After Closing the Loop

The open forward path G_1G_2 has an integrator, so its step ramps off without bound (before); closing the unity-feedback loop turns it into a stable, reference-tracking system (after).

```
figure
step(fwd,0:0.01:5)
hold on
step(T,0:0.01:5)
yline(1,'k:', 'HandleVisibility','off')
hold off
legend('Before (open forward path)', 'After (closed loop)', 'Interpreter','latex', 'FontSize',12, 'Location','northwest')
title('Problem 3: Before vs. After Closing the Loop', 'Interpreter','latex', 'FontSize',16)
ylabel('$y(t)$', 'Interpreter','latex', 'FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter','latex', 'FontSize',20)
```

Problem 3: Before vs. After Closing the Loop



Try it yourself

- In Problem 2, change the target **zeta** to 0.2 and recompute **b**: notice how much less damping the system then needs.
- In Problem 3, remove the integrator (`G2 = tf(5,[1 5])`) and see the steady-state error stop being zero.